

Prove each of the statements in 23-33.

4.6.24. For any integer m and any real number x , if x is

not an integer, then $\lfloor x \rfloor + \lfloor m - x \rfloor = m - 1$

Suppose m is any integer, and x is any real number.

We must show that if x is not also an integer, then

the sum of $\lfloor x \rfloor$ and $\lfloor m - x \rfloor$ is $m - 1$.

$$\forall m \in \mathbb{Z}, x \notin \mathbb{Z} \rightarrow \lfloor x \rfloor + \lfloor m - x \rfloor = m - 1$$

Definition of floor:

$$\lfloor x \rfloor = n \leftrightarrow n \leq x < n + 1$$

Proof:

$$\text{Let } n \in \mathbb{Z} = \lfloor x \rfloor$$

Since n is the floor of x :

$$n \leftrightarrow n \leq x < n + 1$$

But $x \notin \mathbb{Z}$. Therefore $x \neq n$, so:

$$n \leftrightarrow n < x < n + 1$$

$$n + \lfloor m - x \rfloor = m - 1$$

$$\lfloor m - x \rfloor = m - 1 - n$$

But we must show this. Since $m - 1 - n$ is

the floor of $\lfloor m - x \rfloor$:

$$(m - 1 - n) \leftrightarrow (m - 1 - n) \leq m - x < (m - 1 - n) + 1$$

We know $n < x < n + 1$, and we want to find

$m - x$. So we can multiply the inequality

by -1 :

$$-n > -x > -(n + 1)$$

Now we can add m to the inequality:

$$m - n > m - x > m - n - 1$$

$\equiv$

$$m - n - 1 < m - x < m - n$$

Therefore, $\lfloor m - x \rfloor = m - n - 1$ and

$$\lfloor x \rfloor + \lfloor m - x \rfloor = n + (m - n - 1) = m - 1,$$

which is what was to be shown.

□

Solution: If $x \notin \mathbb{Z}$, then $\lfloor x \rfloor + \lfloor m - x \rfloor$ is $m - 1$, for any $m \in \mathbb{Z}$.